

**Features:**

- Enterprise-grade enhanced hybrid cloud support: It supports all major public and private cloud environments including AWS, Azure, GCP, OpenStack, and VMWare vSphere and vCloud.
- Support for entire VMware stack: Cloudify is the only open source orchestration platform supporting the entire VMware stack; all VMware plugins are open source and available in the Cloudify Community edition.
- Public shared images for both AWS and OpenStack: Prebaked Cloudify Manager environments are now available for AWS through a shared AMI, and OpenStack through a QCOW image. This enables simple bootstrapping of a full-fledged Cloudify environment in minutes.
- Deployment update: This allows updating of application deployments, enabling application operations engineers and developers to introduce topology changes and include new resources to run TOSCA deployments.
- In-place manager upgrade: The new Cloudify Manager upgrade process provides fully automated in-place upgrades for all manager infrastructure without any downtime to the managed services; in-place upgrade allows easy migration between Cloudify versions and application of patched versions

Official website: <https://cloudify.co/>

Latest version: 3.4

**ManageIQ**

ManageIQ delivers the insight, control, and automation that enterprises need to address the challenges of managing hybrid IT environments. It allows you to understand the current state of your environment, provide self-service for end users, and enforce compliance policies.

**Features:**

- Insight: Discovery, monitoring, utilisation, performance, reporting, analytics, chargeback, and trending.
- Control: Security, compliance, alerting, policy based resource and configuration management.
- Automate: IT process, task and event, provisioning, workload management and orchestration.
- Integrate: Systems management, tools and processes, event consoles, CMDB, RBA, and Web services.

Official website: <https://www.manageiq.org/>

Latest version: 4.2.0

**OpenNebula**

OpenNebula is a cloud computing platform for managing heterogeneous distributed data centre infrastructures. The OpenNebula platform manages a data centre's virtual infrastructure for private, public and hybrid implementations of Infrastructure as a Service. The two primary uses of the

OpenNebula platform are data centre virtualisation and cloud deployments based on the KVM hypervisor, LXD system containers, and AWS Firecracker microVMs. The platform is also capable of offering the cloud infrastructure necessary to operate a cloud on top of existing VMware infrastructure.

OpenNebula orchestrates storage, network, virtualisation, monitoring, and security technologies to deploy multi-tier services (e.g., compute clusters) as virtual machines on distributed infrastructures, combining both data centre resources and remote cloud resources, according to allocation policies.

**Features:**

- Simple, clean, intuitive GUI for users and admins, with different views
- Easy self-provision of containerised and virtualised workflows from a catalogue
- Fine-grained accounting and monitoring
- Build your private marketplace to share and distribute applications within your organisation
- Dynamic creation of clusters as pools of hosts
- Dynamic creation of virtual data centres as fully-isolated virtual environments
- Federation of multiple zones for scalability, isolation or multiple-site support
- Powerful and flexible scheduler — deploy your workload in different locations
- Powerful user, group and role management
- Integration with enterprise and open source user management services

Official website: <https://opennebula.io/>

Latest version: 6.0 

**References**

- [1] <https://www.forrester.com/report/A+Clear+Multicloud+Strategy+Delivers+Business+Value/-/E-RES128781>
- [2] <https://www.cloudbolt.io/best-cloud-management-platforms-2019/>
- [3] <https://mist.io/>
- [4] <https://cloudify.co/>
- [5] <https://www.manageiq.org/>
- [6] <https://opennebula.io/>

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